



What other factors affect performance?

We saw that the drives that posted comparatively lower scores in the access time test were also the ones that returned low data transfer scores. Access time affects the random access of data, which was evident in the random data transfer tests. The 20 GB Seagate ST20410A posted a score of 16.6 ms in the access time test. The drive that posted very good access time scores was the **60 GB Samsung SV6004H**, which posted a good score of 9.5 ms. Expectedly, this drive exhibited good overall performance in the rest of the data transfer tests.

The CPU utilisation of the hard disk, though neglected by most of us at times, determines the demands that the hard disk places on T processor during its operations. This is especially significant to 5400-rpm drives, as they do not have the headroom in terms of data bandwidth as compared to their 7200-rpm counterparts. These scores were logged by the HDtach benchmark and the drive that posted a good score in this test was the **20 GB Maxtor 2B020H1 541DX**, which logged just 2.6 per cent of CPU utilisation. On the other end of the scale, the drive that utilised more CPU power was the 20 GB Seagate ST20410A, at a very significant 12.7 per cent. This means that the Maxtor drive would be ideally suited for situations when you want to use the hard disk in systems that are powered by an older processor, where any additional strain on it would be very noticeable in terms of overall system performance. The same would be the case in applications where you would be using software that are implicitly demanding on the main processor, such as image, sound and video editing, 3D authoring, etc.

There are two primary parameters that you need to consider when deciding on



Maxtor 536DX: this 100 GB drive gives you all the storage space you need

available for up to Rs 6,000, coupled with very good performance numbers. This makes these drives very well suited to tasks that need a good level of performance while still offering 40 GB capacities.

If you're looking for a greater amount of storage, then the 60 GB Samsung SV6004H hits the sweet spot with a price tag of just Rs 8,000. This was, in fact, the cost of a 40 GB hard disk about four months ago. If you're willing to shell out a few thousand rupees extra, you can even get an 80 GB or a 100 GB hard disk—these cost between Rs 12,000 and Rs 15,500. Of special mention here is the Maxtor 100 GB hard disk (the largest in the country right now), which is available at Rs 12,800. This is definitely a better choice, both in terms of price and performance, over the 80 GB Seagate ST380020A, for example.

How big is big?

When it comes to tasks that require large files to be written to and read from the hard disk, you should opt for one that sports high sequential data transfer speeds such as the 40 GB IBM Deskstar or the 20 GB Seagate Barracuda drives. If, on the other hand, your job involves working with smaller files such as database applications that access information frequently from the drive, or multimedia and Web authoring tools, then the random data transfer speed of the drive would play a pivotal role in the performance and response of your applications. Here, the drive that excelled was the 76.8 GB IBM Deskstar.

Finally, when it comes to applications, you would need to decide the size of the hard disk you're willing to settle for depending on the applications you plan to use. For example, if you're a home user who's looking for a performance hard disk and you plan to use applications such as

intensive gaming, 3D software and sound and video editing, then you would at least need a 20 GB hard disk. Depending upon the volume of files you plan to create, you could even wind up using all of the space on your 40 GB hard disk! (Refer to the table, 'How Much Space is Enough?' highlighting the suggested hard disk sizes in relation to the files and applications you most often work with.) As a rule, text documents and spreadsheet files require the least amount of hard disk space. In the middle range, there would be files such as Power-Point presentations, MP3 audio files, etc.



Finally, if you're a movie buff and store feature-length movies on your hard disk itself, at the rate of about 700 MB per movie, your hard disk could get filled in direct proportion to your interest in watching movies!

Files such as MP3, audio and movies do not really need a high-speed hard disk to work well.

Performance hard disks come into play if you work with audio and video files and are involved in multimedia authoring applications that use large files that are accessed constantly. This would also include Photoshop where average file sizes would range anywhere between 5 and 50 MB. In these cases, a fast hard disk drive will ensure that you don't have to wait long when working with many such files.

CONCLUSION

From this roundup, it is apparent that the array of hard disks available are so wide in terms of capacity,



IBM Deskstar 60GXP: the performance hero